

Title

Bactericidal Activity and Tolerability of Novel Combination Therapy in Drug-Resistant Tuberculosis: A Pharmacodynamic Investigation

Abstract

Patients with rifampicin-resistant tuberculosis (RR-TB) require intensive multi-drug regimens. We assessed the bactericidal effectiveness and safety profile of a four-drug combination therapy compared with conventional treatment in a cohort of RR-TB patients enrolled across two treatment centers. Participants received the investigational regimen as part of a treatment protocol implemented in clinical settings in Eastern Europe and South Africa. Sputum smear conversion and bacterial load reduction were measured at defined intervals. The investigational combination achieved approximately 18% higher bacillary clearance compared to standard therapy in the overall population, with particularly strong results in patients without severe systemic disease. However, accelerated bacterial reduction was observed in only a subset of patients. Hematologic complications, including anemia and reduced neutrophil counts, were more prevalent among those receiving higher cumulative drug exposure, though these did not necessitate treatment discontinuation in most cases. The clinical findings support the use of this intensive multi-drug approach as a viable treatment option for patients with resistance to first-line agents.